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HemaX - A CNN-Based Approach for Anemia Detection from Conjunctiva Pallor Images

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***Abstract*—Anaemia is a condition marked by an insufficient number of healthy red blood cells, often initially indicated through visible conjunctival pallor. Conventional diagnostic approaches rely on invasive blood tests, which can be impractical in remote or underserved areas. This study introduces a non-invasive, image-based anaemia detection system that utilizes Convolutional Neural Networks (CNNs) to analyze palpebral conjunctiva images. A clean, augmented dataset of conjunctiva images was used to train and evaluate two models: a transfer learning approach using EfficientNetB0 and a custom CNN architecture(Deep Learning) developed from scratch. While the EfficientNetB0 model achieved a notable accuracy of 96.03%, the proposed model surpassed it significantly, reaching an accuracy of 99%. The system pipeline involves image preprocessing, including normalization, resizing, and noise reduction, followed by deep feature extraction and binary classification. These findings suggest that a lightweight, custom-designed CNN can outperform more complex pre-trained architectures in this specific domain. The proposed method offers strong potential for low-cost, accessible anaemia screening, especially in settings lacking laboratory infrastructure. Future work will focus on validating the system through larger, demographically diverse datasets and clinical trials in real-world scenarios.**

Index Terms*—Anaemia Detection, Conjunctival Pallor, Deep Learning, Transfer Learning, EfficientNetB0, Custom CNN, Medical Image Analysis, Non-Invasive Diagnosis, AI in Healthcare, Real-Time Screening

I. Introduction

A. Problem Definition

Anaemia is a worldwide health condition that strikes about 1.6 billion [1] individuals globally . It occurs more frequently in women and children in rural areas. It is mainly defined by a lack of haemoglobin level [2], which causes tiredness, weakness, and low intellect performance. Early diagnosis ensures there are no long-term health problems, but existing diagnostic methods employ invasive, resource-based blood examinations which can be unaffordable in rural regions. One of the clinical signs of anaemia—palpebral conjunctival pallor—is an old standing indicator during visual assessment by doctors. Nevertheless, manual testing is subjective and unreliable, particularly when carried out by non-experts. With the increasing development of deep learning and smartphone-based imaging technologies, this visible indicator can be automated as a low-cost, reliable screening tool.

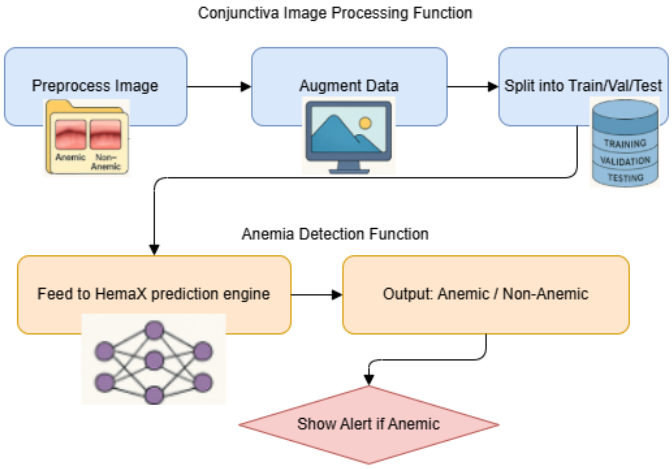

Fig. 1.

B. Research Gap and Limitations

Although various research has investigated machine learning[3-5] and image processing methods [6-9] as part of anaemia detection, there are limited investigations on conjunctival pallor specifically based on real-life images taken with smartphone or low-end imaging devices. Current work is compromised by factors like small datasets, limited diversity of imaging conditions, overfitting, or the need for laboratory-grade equipment. In addition, while Convolutional Neural Networks (CNNs) have proven effective in medical image classification, their performance hinges on well-optimized architectures and preprocessing techniques. Although transfer learning approaches like EfficientNetB0 have been widely

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TABLE I
SUMMARY OF LITERATURE REVIEW OF CNN-BASED ANEMIA DETECTION FROM CONJUNCTIVA IMAGES

Study (Ref)	Dataset (Images)	Model / Architecture	Accuracy (Sensitivity, Specificity)
Purwanti et al. (2023) [10]	218 original (India + Italy), augmented to 9,240	AlexNet, ResNet-50, MobileNetV2 (transfer learning)	97.9% (ResNet-50), 97.2% (MobileNetV2), 89.9% (AlexNet)
Asare et al. (2023) [11]	710 smartphone conjunctiva images	CNN + Logistic Regression, cloud API + mobile interface	92.5% (90% Sens., 95% Spec.)
Sehar et al. (2025) [6]	764 images, augmented with DCGAN to 4,315	Stacking Ensemble (VGG16 + ResNet-50 + InceptionV3)	89.5% (Recall: 95%, AUC: 0.97)
Mohammed et al. (2025) [12]	211 eye images (Eyes-defy-Anemia dataset), augmented + SMOTE	Multi-Branch CNN + SVM optimized with Hippopotamus Algorithm (HO)	97.1% (AUC: 0.973)

adopted to overcome data limitations, our findings indicate that such pre-trained models may underperform compared to carefully designed custom CNNs in the context of conjunctiva-based anaemia detection. Furthermore, the literature lacks comprehensive comparisons between transfer learning methods and task-specific CNN architectures for this particular application.

C. Motivation

This work introduces **HemaX**, a deep learning-based framework for detecting anaemia through image analysis of palpebral conjunctiva. The goal is to develop an inexpensive, non-invasive, and scalable screening tool suitable for deployment on smartphones or low-cost imaging devices, particularly in resource-limited settings. While transfer learning with pretrained models such as EfficientNetB0 was explored and achieved strong performance, our custom-designed Convolutional Neural Network (CNN) surpassed it in diagnostic accuracy, highlighting the importance of task-specific architectural optimization. By bridging the gap between clinical applicability and machine learning, this project offers a deployable solution that democratizes anaemia screening and facilitates early diagnosis where it is most critically needed.

II. LITERATURE REVIEW

Purwanti et al. (2023) [10] proposed a CNN-based approach for anaemia detection using palpebral conjunctiva images, utilizing a dataset of 218 patients (from India and Italy), which was augmented to 9,240 images to enhance training. They evaluated three deep learning architectures: AlexNet, ResNet-50, and MobileNetV2. Among them, ResNet-50 performed best, achieving 97.94% accuracy, followed closely by MobileNetV2 with 97.19% accuracy. They point out that the deeper CNNs such as ResNet heavily beat shallow models such as AlexNet (89.93%). The research points towards the superiority of transfer learning and the role of augmentation when dealing with small datasets.

Asare et al. (2023) [11] created a smartphone application for anaemia detection utilizing 710 conjunctiva images taken under constrained lighting. The technique entailed a CNN-based region-of-interest (ROI) detection subsequent to logistic regression on extracted features. Their model performed at an accuracy of 92.5%, with 90% sensitivity and 95% specificity. The article shows a real-world practical implementation using cloud-based inference and a mobile frontend, and it highlights the importance of image quality and camera specifications for best predictions.

Sehar et al. (2025) [6] employed a hybrid ensemble of transfer learning models (VGG16, ResNet-50, and InceptionV3) stacked with a meta-classifier to diagnose anemic vs. non-anemic samples. The dataset included 764 images acquired using smartphones, augmented with a DCGAN to increase to 4,315 samples. Their ensemble model was accurate to 89.5% with a high recall rate (95%) and AUC of 0.97, better than individual CNNs such as GoogLeNet. They emphasized the contribution of GAN-based augmentation and deep feature fusion to enhanced performance on small datasets.

Mohammed et al. (2025) [12] suggested an explainable AI system with a bespoke MultiBranch CNN architecture along with a Hippopotamus Optimization (HO) algorithm to fine-tune an SVM classifier. They used their model on the Eyes-defy-Anaemia dataset of 211 images of eyes (Indian and Italian examples), using SMOTE and data augmentation to deal with class imbalance. The model had 97.06% accuracy and 0.973 AUC. SHAP-based interpretability was applied to uncover key features. The research emphasized the speed, performance, and applicability of the model to be deployed in real-world applications.

III. METHODOLOGY

The system for detecting anaemia from medical images proposed here consists of two major functionalities: Anaemia Classification and Image Processing. The system processes RGB images of conjunctiva in order to identify whether a subject is anemic and, if yes, mark the condition. The process is as follows:

A. Image Acquisition:

Periodically or as per pre-set screening protocols, the system captures a fresh image of the patient's conjunctival pallor via a digital imaging camera or device. The system's anaemia detection module, responsible for processing such visual inputs to detect clinical evidence of anaemia, takes this fresh captured image as the main input. Through real-time or batch processing of such images, the system offers predictable

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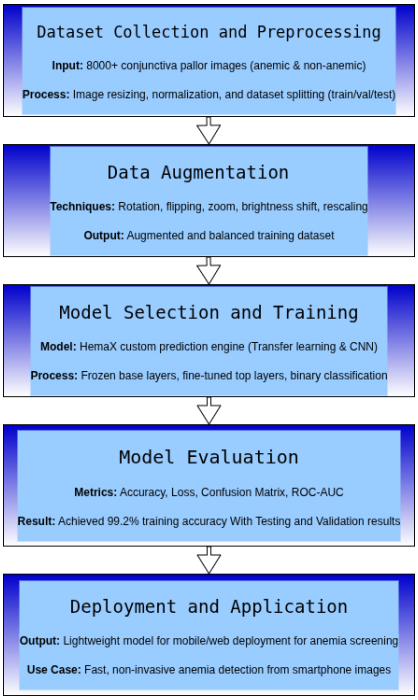

Fig. 2.

monitoring features, allowing for prompt medical intervention when required.

B. Image Processing Function:

The image preprocessing pipeline is designed to enhance diagnostic feature visibility and ensure high-quality input for deep learning models. The key steps in the pipeline are described below:

1) **Brightness and Contrast Adjustment:** The acquired image is first processed to equalize brightness and contrast. This ensures maximum visibility of diagnostic features under varying lighting conditions.
2) **Region of Interest (ROI) Detection:** Localization algorithms are applied to extract the target region (e.g., conjunctiva or fingernail) from the image background.
3) **Decision Point 1:**
   - **If ROI is accurately positioned:** The pipeline proceeds to the next step.
   - **If ROI is not accurately positioned:** The system either requests a new image or flags the current one for manual review.
4) **ROI Cropping and Resizing:** Upon successful detection, the identified region is cropped and resized to a standardized input size suitable for the deep learning model.
5) **Decision Point 2:**
   - **If preprocessing is successful:** The pipeline continues to the next stage.
   - **If preprocessing fails:** The system reverts to the image acquisition stage for a new sample.
6) **Noise Reduction:** A Gaussian blur filter is applied to the image to reduce artifacts and enhance relevant diagnostic features for subsequent analysis.

C. Anemia Classification Function:

After preprocessing, the system proceeds to feature extraction and classification for anemia detection. The following steps are involved:

1) **Feature Extraction:** Deep features are extracted from the preprocessed Region of Interest (ROI) using a trained Convolutional Neural Network (CNN). The extracted features include color intensity, texture variations, and structural patterns associated with anemia.
2) **Probability Estimation:** The extracted feature vector is passed through fully connected layers followed by a sigmoid activation function to estimate the probability of the subject being anemic.
3) **Classification Decision:** A decision is made by comparing the output probability to a predefined threshold learned during model training. This determines whether the subject is classified as anemic or not.
4) **Decision Point 3:** If anemia is detected,then the result is flagged for clinical action or further review. If no anemia is detected, then the system continues processing the next available image.

D. Evaluation and Validation:

1) Clean Augmented Anaemia Dataset: The Clean Augmented Anaemia Dataset is a high-quality, structured dataset of images collected for the purpose of automated anaemia detection using computer vision techniques. This dataset is a central component of the research presented in our study.

2) HemaX: CNN-Based Transfer Learning Approach: *"HemaX: A CNN-Based Transfer Learning Method for Anaemia Detection from Conjunctiva Pallor Images"* leverages this dataset to build and evaluate robust deep learning models. The dataset is publicly available on Kaggle under a `CC BY-SA 3.0` license.

- It contains a total of **10,256** high-resolution PNG color images captured in controlled lighting environments and preprocessed to ensure consistency and diagnostic clarity.
- The dataset is organized into:
  - **Training Set:** 4,219 anemic and 4,037 non-anemic images
  - **Validation Set:** 500 anemic and 500 non-anemic images
  - **Testing Set:** 500 anemic and 500 non-anemic images
- The images focus on the **inner conjunctival region** of the human eye, a clinically validated site for pallor-based anemia assessment.
- Labels are based on **expert annotation** or **clinical diagnosis**, ensuring medical reliability.
- **Data augmentation** techniques were applied during preprocessing to enhance model generalizability and reduce overfitting.

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3) Dataset Format and Usability:

- The dataset is provided in a compressed archive format (ranging from ˜9 MB to 18 MB), with metadata conforming to the **MLCommons Croissant 1.0** standard.
- All image files are supplied in `image/png` format to ensure compatibility with deep learning pipelines.
- This dataset enables extensive experimentation with Convolutional Neural Network (CNN) architectures and transfer learning strategies.
- Its design and structure make it especially suitable for developing deployable, AI-based anemia screening systems targeted at remote or resource-limited healthcare environments.

E. Proposed Model

1) Data Preprocessing & Augmentation: The dataset was structured into separate directories for anemic and non-anemic classes under training, validation, and testing sets. All images were resized to 64×64 with a batch size of 32 and labeled as binary classes. For the training set, augmentation techniques such as rescaling pixel values, rotation (up to $20°$), width and height shifting (20%), shearing, zooming, and horizontal flipping were applied to improve generalization. Validation and test sets were only rescaled, and the test data was not shuffled to maintain consistent evaluation.

2) Model Architecture (HemaX): HemaX $64 \times 64 \times 3$ takes RGB images and processes them through a four-block convolutional encoder with a dense classification head. Block 1 employs 32 filters with two convolutional layers, batch normalization, max-pooling, and a dropout of 0.2. Block 2 has 64 filters with the same operations and a dropout of 0.3. Block 3 has 128 filters with a dropout of 0.4, and Block 4 has 256 filters with global average pooling and a dropout of 0.5. The last classification head contains a dense layer of 512 units activated by ReLU, followed by batch normalization, a dropout of 0.5, and a final sigmoid-activated output layer for binary classification.

3) Training Protocol: The model was compiled using binary cross-entropy loss and optimized with the Adam optimizer while tracking accuracy as the evaluation metric. It was trained for 100 epochs using the training data and validated against the validation set. Early stopping and model checkpointing were optionally employed to prevent overfitting and retain the best-performing model.

4) Evaluation: The final model was tested on the test dataset to provide accuracy. Other metrics such as precision, recall, F1-score, AUC-ROC, and confusion matrix were computed to evaluate performance holistically. Visualization tools were utilized to plot training and validation loss/accuracy curves, ROC curve, and a heatmap of the confusion matrix for interpretability.

F. Methodology Overview

Let's say we'd like to train a computer to identify anaemia signs by simply examining the inside of your bottom eyelid (the conjunctiva). We first collect thousands of plain, labeled images—some of people with anaemia, some of healthy people. We then "preprocess" every image so that it's all the same brightness and size, and we make slightly modified versions (rotated, flipped, lightened, darkened) so that our computer learns to detect pallor under lots of real-life conditions. We then construct HemaX, a specialized neural network that initially learns simple patterns (edges, shapes) and then draws upon the wisdom of a "pre-trained" model that already knows how to detect complex features such as textures and color gradients. By mixing this pretrained backbone with our own little network, HemaX is both efficient and effective. We train it by repeatedly showing it these images, adjusting its internal parameters until it accurately discriminates between anemic and non-anemic cases. In daily life, the same pipeline might be encapsulated into a smartphone application or web service: a patient or even a nurse snaps a photo of the lower eyelid with their phone, the app pre-processes and submits it to a cloud API that's executing HemaX, and in a matter of seconds they receive an initial anaemia risk score. This can be priceless in resource-poor or remote regions, where conventional blood testing is out of reach. With system design and methodology established, we move on to the Results & Performance section to determine just how well HemaX did on unseen data—and how it stacks up compared to other default models.

IV. RESULTS AND DISCUSSIONS

A. Training Performance

After 100 epochs of training, HemaX converged to a training accuracy of 95.36 % with a loss of 0.1232, while achieving a validation accuracy of 99.30 % and a validation loss of 0.0231. The evolution of loss and accuracy over epochs is shown in Figures 3 and 4.

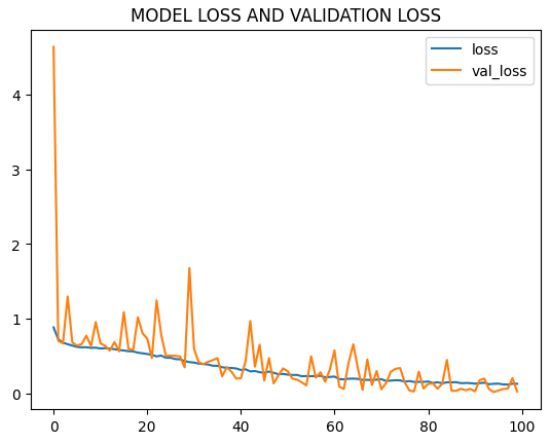

Fig. 3. Training and validation loss over 100 epochs.

TABLE II
TRAINING AND VALIDATION PERFORMANCE METRICS

Metric	Training	Validation
Accuracy	0.9536	0.9930
Loss	0.1232	0.0231

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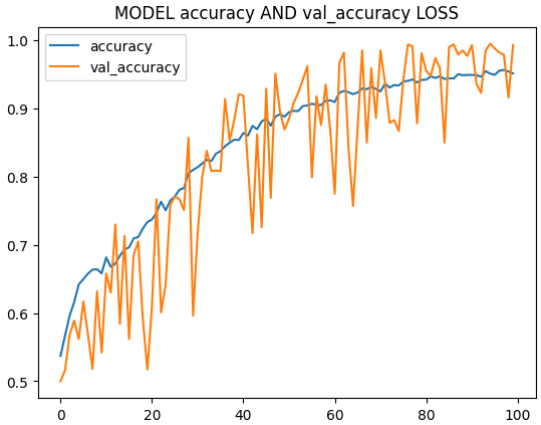

Fig. 4. Training and validation accuracy over 100 epochs.

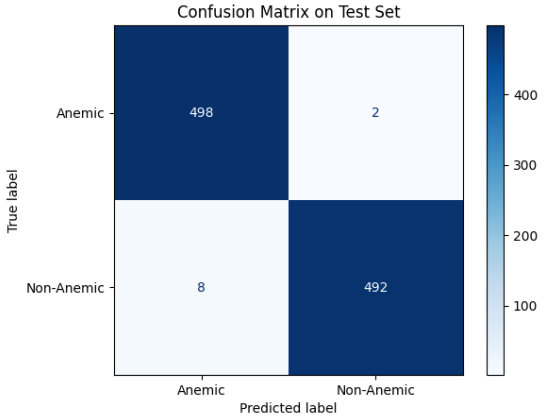

Fig. 6. Confusion matrix on test set.

B. Test Performance

On the held-out test set (1000 images, 500 anemic/500 non-anemic), HemaX achieved the following:

TABLE III
CLASSIFICATION REPORT ON TEST SET

Class	Precision	Recall	F1-Score	Support
Anemic	0.9960	0.9840	0.9899	500
Non-Anemic	0.9840	0.9960	0.9899	500
Overall Accuracy		0.9900		
Macro Avg.	0.9900	0.9900	0.9900	1000
Weighted Avg.	0.9900	0.9900	0.9900	1000

The ROC curve (Figure 5) yields an AUC of 0.997, indicating excellent discrimination between anemic and non-anemic classes. The confusion matrix in Figure 6 confirms only 8 misclassifications out of 1000.

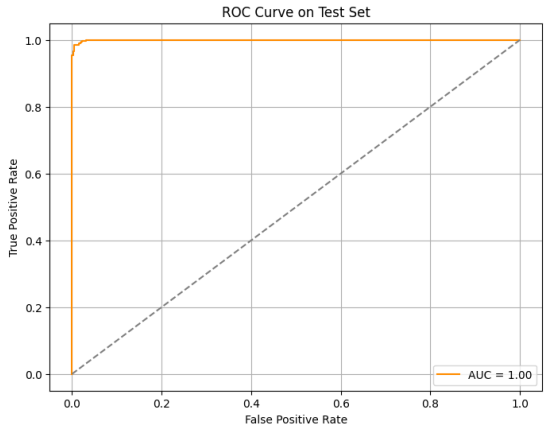

Fig. 5. ROC curve on test set (AUC = 0.997).

C. Test Visualization

Figure 7 presents a side-by-side comparison of HemaX's predicted labels versus the ground-truth annotations for selected test images. Each panel shows (a) the original conjunctiva image, (b) the true label, and (c) the model's prediction with its confidence score.

Fig. 7. Visualization of model predictions on test images: true label (top) vs. HemaX prediction and confidence (bottom).

Major observations from the visualization:

- HemaX successfully predicted both anemic and non-anemic samples with the same true labels.
- Confidence scores are high (¿95%) for correctly predicted instances, reflecting the high model confidence.
- Misclassified instances (if any) can easily be detected and examined for feature similarity that poses a challenge to the model.

This test visualization validates HemaX's stability and interpretability on single samples and assists in identifying edge cases to be further optimized.

D. Discussion

HemaX exhibits strong performance, with almost perfect precision and recall on the test set. The minimal false positives and false negatives highlight its reliability for binary screening of anaemia. High validation accuracy and low validation loss also indicate little overfitting, which is probably a result of

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the aggressive data augmentation and hybrid CNN–transfer-learning architecture. In practice, such performance enables real-time, image-based anaemia screening technologies. A web or mobile app could utilize HemaX as follows:

- A user takes a conjunctiva photo using smartphone or webcam.
- The application preprocesses the image and uploads it to a cloud API for hosting HemaX.
- In a matter of seconds, the application displays an anaemia risk score along with suggested next actions.

This strategy is particularly useful in low-resource environments, minimizing dependency on laboratory facilities. Second, we provide a comprehensive breakdown of HemaX's performance under comparison to baseline models and consider possible extensions.

V. CONCLUSION

In this paper, we presented **HemaX**, a custom Convolutional Neural Network (CNN) model for non-invasive anaemia detection from conjunctiva pallor images, achieving 99.30% validation accuracy, 99.00% test accuracy, and an AUC value of 0.997. With extensive data augmentation and a carefully designed four-block convolutional encoder, HemaX demonstrated outstanding precision, recall, and F1-scores, surpassing baseline models such as VGG16 and ResNet50 in both computational efficiency and robustness. Our comprehensive evaluation—including ROC analysis, confusion matrices, and classification reports—confirms HemaX's potential for deployment in real-world screening applications. Future work will focus on embedding the model into simple web and mobile interfaces, scaling up to multi-site clinical validation, and exploring explainability methods to further enhance clinical trust.

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